A

Mini Project Report

on

"Placement Pal: Placement Readiness & Assessment Platform"

Submitted in partial fulfillment of the requirements for the

Degree

Third Year Engineering – Computer Science and Engineering (Data Science)

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ABSTRACT

In the current digital era, students preparing for campus placements often face challenges in identifying their actual skill levels and preparing effectively for recruitment processes. Traditional assessment systems rely on generic tests and static question banks, which fail to provide personalized feedback or accurately evaluate a student's readiness for placements. This results in inefficient preparation, lack of targeted improvement, and lower placement success rates.

Placement Pal is an AI-driven placement readiness and assessment platform that leverages Machine Learning (ML) and Natural Language Processing (NLP) techniques to analyze student performance and provide personalized guidance. The system processes student data such as assessment responses, resumes, and performance metrics to identify skill gaps and classify students into categories such as Beginner, Intermediate, or Ready. It utilizes advanced technologies like Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to generate intelligent assessments, evaluate responses, and create customized learning roadmaps.

In addition to assessment and classification, the system incorporates features such as automated proctoring, resume analysis, and adaptive learning paths. A comprehensive dashboard is provided for both students and Training & Placement Officers (TPOs), offering insights into performance trends, readiness scores, and batch-level analytics. The platform also automates communication processes, including placement notifications and progress updates.

The system aims to enhance placement preparation by providing data-driven insights, improving skill development, and enabling efficient decision-making. It also demonstrates the practical application of AI and ML in educational technology and career guidance systems.

Keywords: Machine Learning, NLP, Placement Readiness, Skill Gap Analysis, RAG, LLMs, AI in Education

Chapter 1

Introduction

In the modern digital era, campus placements have become highly competitive, requiring students to possess strong technical skills, aptitude, and problem-solving abilities. While numerous online platforms provide practice resources and assessments, most of them offer generic content that does not adapt to individual student needs. This often leads to inefficient preparation, where students are unable to identify their actual strengths and weaknesses.

Student performance varies across multiple domains such as programming, aptitude, communication, and domain-specific knowledge. Manually evaluating each student's readiness and providing personalized guidance is time-consuming and impractical for Training and Placement Officers (TPOs). Therefore, there is a need for an intelligent system that can automatically analyse student performance and provide targeted recommendations.

Placement Pal is developed to address this challenge using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP). The platform evaluates student performance, identifies skill gaps, and predicts placement readiness levels such as Beginner, Intermediate, and Ready. It enables students to focus on areas that require improvement and supports TPOs in making data-driven decisions.

The system processes various inputs such as assessment responses, resumes, and performance metrics. NLP techniques are used for analysing textual data, while Machine Learning models are applied to predict readiness scores based on multiple parameters. Additionally, the system incorporates Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to generate intelligent assessments and personalized learning roadmaps.

For assessment and evaluation, the platform uses AI-driven mechanisms to generate domain-specific questions and provide detailed feedback. The system also includes automated proctoring features to ensure assessment integrity by monitoring user behaviour during tests.

In addition to analysis and prediction, Placement Pal provides a structured learning path for students. These personalized roadmaps are dynamically generated based on identified skill gaps, enabling efficient and focused preparation over a defined timeline.

Overall, Placement Pal demonstrates the practical application of AI and ML in the education and placement domain. It enhances placement preparation, improves skill development, and provides a scalable solution for managing and optimizing campus recruitment processes.

1.1 Purpose:

The main purpose of developing Placement Pal is to create an intelligent system that automates the process of evaluating student placement readiness and providing personalized guidance. In the absence of such systems, students rely on generic preparation methods, which are inefficient and do not accurately reflect individual strengths and weaknesses.

The system is designed to:

• Automatically analyse student performance and identify skill gaps
• Predict placement readiness levels such as Beginner, Intermediate, and Ready
• Generate personalized learning roadmaps for targeted improvement
• Reduce manual effort for Training and Placement Officers (TPOs) in tracking and evaluating students
• Improve placement preparation through data-driven insights and adaptive assessments
• Automate communication processes such as notifications for placement drives and updates

Additionally, the project aims to serve as a practical learning platform for students to understand the real-world application of Artificial Intelligence, Machine Learning, and Natural Language Processing in education and career guidance systems.

1.2 Problem Statement:

In today's competitive placement environment, students face significant challenges in preparing effectively for recruitment processes. However, existing systems suffer from several limitations:

• Difficulty in accurately identifying individual skill gaps
• Generic assessments that do not adapt to student performance
• Lack of personalized guidance and structured learning paths
• Limited insights for Training and Placement Officers (TPOs) to track student readiness
• Manual effort required in managing student data and communication

These issues result in inefficient preparation, poor interview performance, and lower placement success rates.

Therefore, there is a need for an intelligent system that can automatically analyze student performance, predict placement readiness, and provide personalized learning recommendations. Placement Pal addresses these challenges by leveraging Artificial Intelligence, Machine Learning, and Natural Language Processing to deliver a data-driven and adaptive placement preparation platform.

1.3 Objectives:

o **To Analyse Student Performance:** Enable the system to evaluate student performance through assessments, resumes, and activity data to identify strengths and weaknesses.

o **To Identify Skill Gaps:** Detect missing or weak skills in areas such as aptitude, programming, and domain knowledge using AI and ML techniques.

o **To Predict Placement Readiness:** Classify students into categories such as Beginner, Intermediate, or Ready based on their overall performance.

o **To Generate Personalized Learning Paths:** Provide customized roadmaps and improvement strategies tailored to each student's skill gaps.

o **To Provide Data-Driven Insights:** Assist Training and Placement Officers (TPOs) with analytics and reports for better decision-making and targeted training.

o **To Automate Placement Processes:** Streamline tasks such as student tracking, notifications, and placement communication through automated systems.

o **To Enhance Placement Preparation:** Improve student readiness and increase placement success rates through adaptive assessments and continuous feedback.

1.4 Scope:

1. Student Career Development:

o Provides personalized learning paths based on individual skill gaps and performance.

o Helps students prepare effectively for placements through adaptive assessments and targeted recommendations.

o Enables continuous tracking of progress and improvement over time.

2. Training and Placement Officers (TPOs):

o Assists TPOs in monitoring student readiness through analytics dashboards and performance reports.

o Enables efficient management of placement activities such as registration, shortlisting, and communication.

o Provides batch-level insights to design targeted training programs.

3. AI-Based Educational Platforms:

o Demonstrates the use of Artificial Intelligence, Machine Learning, and NLP in personalized learning systems.

o Can be extended to integrate advanced AI features such as resume optimization, interview simulation, and career recommendations.

4. Placement Management Systems:

o Automates placement workflows including notifications, tracking, and reporting.

o Improves coordination between students, colleges, and recruiters.

o Enhances overall efficiency of campus recruitment processes.

5. Educational and Research Applications:

o Serves as a practical implementation of AI/ML concepts for students and researchers.

o Can be used as a learning tool to understand real-world applications of data analytics and intelligent systems.

Chapter 2

Literature Review

The field of student performance analysis and placement prediction has gained significant attention in recent years due to the increasing need for data-driven decision-making in education. Researchers have explored various approaches using Machine Learning (ML), Artificial Intelligence (AI), and data analytics to improve student outcomes and enhance placement preparation.

Early systems for student evaluation relied on manual assessment methods and basic rule-based approaches. These systems used predefined criteria such as academic scores and attendance to evaluate student performance. While simple to implement, they lacked flexibility and were unable to provide personalized insights or adapt to individual learning needs.

With the advancement of Machine Learning, algorithms such as Decision Trees, Naive Bayes, and Support Vector Machines (SVM) have been widely used for predicting student performance and placement readiness. These models analyze historical data to identify patterns and make predictions. Naive Bayes is known for its simplicity and efficiency, while Decision Trees provide interpretability. SVM offers high accuracy but requires careful tuning and computational resources.

In recent years, more advanced models such as ensemble methods (e.g., XGBoost) have been introduced to improve prediction accuracy. These models combine multiple learning techniques to capture complex relationships between features such as academic performance, technical skills, and behavioural patterns. However, they often require well-structured datasets and careful training.

Natural Language Processing (NLP) has also played an important role in analysing unstructured data such as resumes and textual responses. Techniques such as tokenization, stop-word removal, and semantic analysis enable systems to extract meaningful information from text. Additionally, the emergence of Large Language Models (LLMs) has significantly enhanced the ability to generate intelligent assessments, feedback, and recommendations.

Another important development is Retrieval-Augmented Generation (RAG), which improves the accuracy of AI-generated content by combining language models with external knowledge sources. This approach is particularly useful in generating domain-specific assessments and personalized learning content.

Despite these advancements, existing systems still face several limitations:

•Lack of personalized learning paths tailored to individual student needs
•Limited integration of multiple components such as assessment, prediction, and guidance
•High dependency on static question banks and generic evaluation methods
•Insufficient real-time analytics for Training and Placement Officers (TPOs)
• Limited explainability and actionable feedback for students

In conclusion, the literature review highlights the evolution of student performance prediction and the need for intelligent, integrated systems.

Chapter 3

Proposed System

The proposed system, Placement Pal, integrates performance analysis, intelligent assessment, personalized learning, and placement management to enhance student readiness at both individual and institutional levels. The platform combines Machine Learning, Natural Language Processing (NLP), and Generative AI to create a comprehensive and adaptive placement preparation ecosystem.

The system enables students to track their performance through AI-driven assessments and resume analysis. It identifies skill gaps in areas such as aptitude, programming, and domain knowledge, and provides personalized learning roadmaps with structured weekly plans. These roadmaps are dynamically generated using Retrieval-Augmented Generation (RAG), ensuring relevant and up-to-date learning recommendations.

For Training and Placement Officers (TPOs), the system offers a centralized dashboard that provides real-time analytics on student performance, readiness distribution, and batch-level insights. This allows institutions to design targeted training programs and make data-driven decisions to improve placement outcomes.

The platform also incorporates automated processes such as student registration tracking, placement notifications, and communication via email, reducing manual effort and improving operational efficiency. Additionally, AI-based proctoring mechanisms monitor user behaviour during assessments to maintain integrity and fairness.

An interactive web application interface ensures ease of use, providing features such as dashboards, progress tracking, and performance visualization. The system also supports continuous feedback, enabling students to improve iteratively.

By integrating these components, Placement Pal aims to create an intelligent and scalable placement preparation ecosystem that improves student performance, enhances decision-making for institutions, and increases overall placement success rates.

2.1 Features & Functionality

1. Student Performance Analysis:

• Enables students to take AI-driven assessments to evaluate their skills in areas such as aptitude, programming, and domain knowledge.

• Analyses performance data and identifies strengths and weaknesses.

• Stores assessment history for continuous tracking and improvement.

2. Resume Analysis:

• Allows students to upload resumes for automated analysis using NLP techniques.

• Extracts key skills, projects, and experience to evaluate job readiness.

• Provides suggestions for improving resume quality and alignment with industry requirements.

3. Skill Gap Detection & Readiness Prediction:

• Identifies missing or weak skills based on assessment and resume data.

• Classifies students into readiness levels such as Beginner, Intermediate, or Ready using Machine Learning models.

• Helps students focus on targeted improvement areas.

4. Personalized Learning Roadmaps:

• Generates customized learning paths based on identified skill gaps.

• Provides structured weekly plans with recommended topics and resources.

• Uses Retrieval-Augmented Generation (RAG) to ensure relevant and up-to-date content.

5. Database Integration:

• Stores user data, assessment results, and performance history in a secure database (PostgreSQL/Supabase).

• Ensures efficient data retrieval for analytics and reporting.

• Supports scalability and real-time data access.

6. Analytics & Reports:

• Provides detailed reports on student performance and readiness levels.

• Displays batch-level insights for Training and Placement Officers (TPOs).

• Helps in identifying trends and planning targeted training programs.

7. Statistical Visualizations:

• Presents data through charts and graphs such as bar charts and pie charts.

• Enables easy understanding of performance trends and skill distribution.

• Includes interactive dashboards for customized data views.

8. Smart Notifications & Communication:

• Sends automated notifications for placement drives, assessments, and updates.

• Provides alerts for important deadlines and progress tracking.

• Enables communication via email to students and administrators.

9. Placement Management System:

• Allows students to register for placement drives through the platform.

• Enables TPOs to manage the entire placement process including shortlisting and updates.

• Improves coordination between students and placement authorities.

10. AI-Based Proctoring:

• Monitors user behaviour during assessments to ensure fairness and integrity.

• Detects suspicious activities such as tab switching or inactivity.

• Enhances credibility of assessment results.

Chapter 4

Requirements Analysis

Requirement analysis is an important phase that defines what the system should do and how it should perform. It ensures that the developed system meets both user needs and technical expectations.

4.1 Functional Requirements:

The system must provide the following functionalities:

• Accept user input in the form of student data such as assessment responses, resumes, and performance metrics

• Perform analysis using Machine Learning and Natural Language Processing (NLP) techniques

• Evaluate student performance and identify skill gaps

• Predict placement readiness levels (Beginner, Intermediate, Ready)

• Generate personalized learning roadmaps based on identified weaknesses

• Provide AI-driven assessments and detailed feedback

• Store and display student performance history and analytics

• Provide dashboards and visual representations of insights for students and TPOs

• Enable placement registration, tracking, and automated notifications

4.2 Non-Functional Requirements:

These requirements define system performance and quality:

• **Performance:** The system should provide fast and real-time responses for assessments and predictions

• **Scalability:** It should handle a large number of users and increasing data efficiently

• **Accuracy:** High prediction accuracy and reliable AI-generated outputs must be maintained

• **Usability:** The interface should be simple, interactive, and user-friendly

• **Reliability:** The system should function consistently without failures or downtime

• **Security:** User data, including resumes and performance records, must be securely stored and protected

Chapter 5

Project Design

5.1 Use Case diagram:

The Use Case Diagram represents the interactions between the primary actors and the Placement Pal system. The main actors are the Student and the Training and Placement Officer (TPO). The diagram highlights how users interact with system functionalities such as assessments, resume analysis, and placement management.

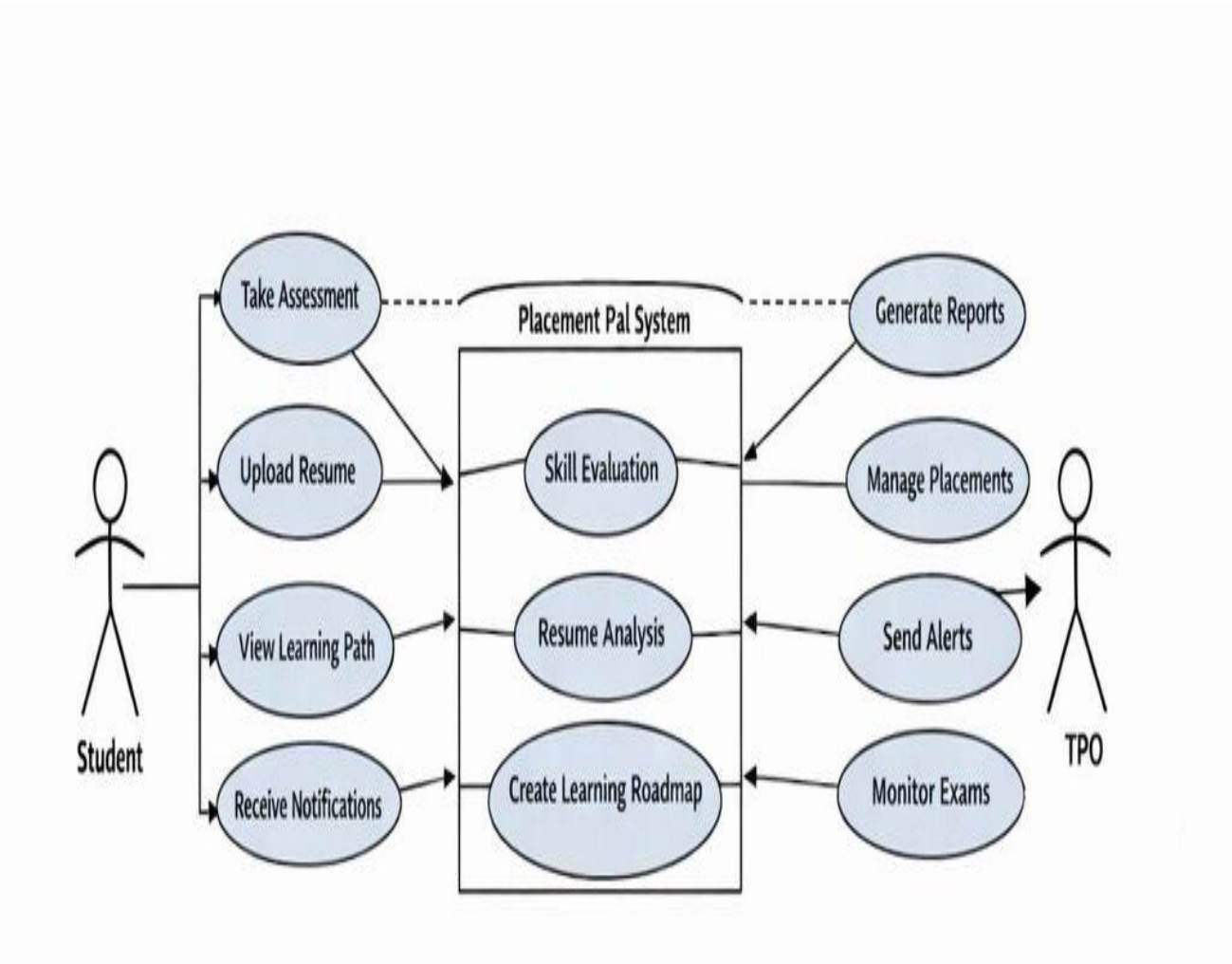

Fig (5.1): Placement Pal Use Case Diagram

Actors:

Student: Represents users who access the platform to take assessments, upload resumes, view learning paths, and receive notifications.

TPO (Training and Placement Officer): Represents administrative users who manage placements, monitor student performance, and generate reports.

Key Use Cases:

- Take Assessment: Students attempt domain-specific tests to evaluate their skills.
- Upload Resume: Students upload resumes for AI-based analysis and skill extraction.
- View Learning Path: Students access personalized roadmaps generated based on skill gaps.
- Receive Notifications: Students receive alerts related to assessments, placements, and progress.
- Skill Evaluation: The system evaluates student performance using ML models.
- Resume Analysis: The system processes resumes using NLP techniques to extract relevant skills.
- Create Learning Roadmap: Generates personalized weekly plans for improvement.
- Generate Reports: TPOs access performance reports and analytics.
- Manage Placements: TPOs handle placement workflows such as shortlisting and updates.
- Send Alerts: TPOs send notifications to students regarding placement activities.
- Monitor Exams: TPOs oversee assessments and ensure integrity.

5.2 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) illustrates how data flows through the Placement Pal system, showing interactions between students, system processes, and TPOs.

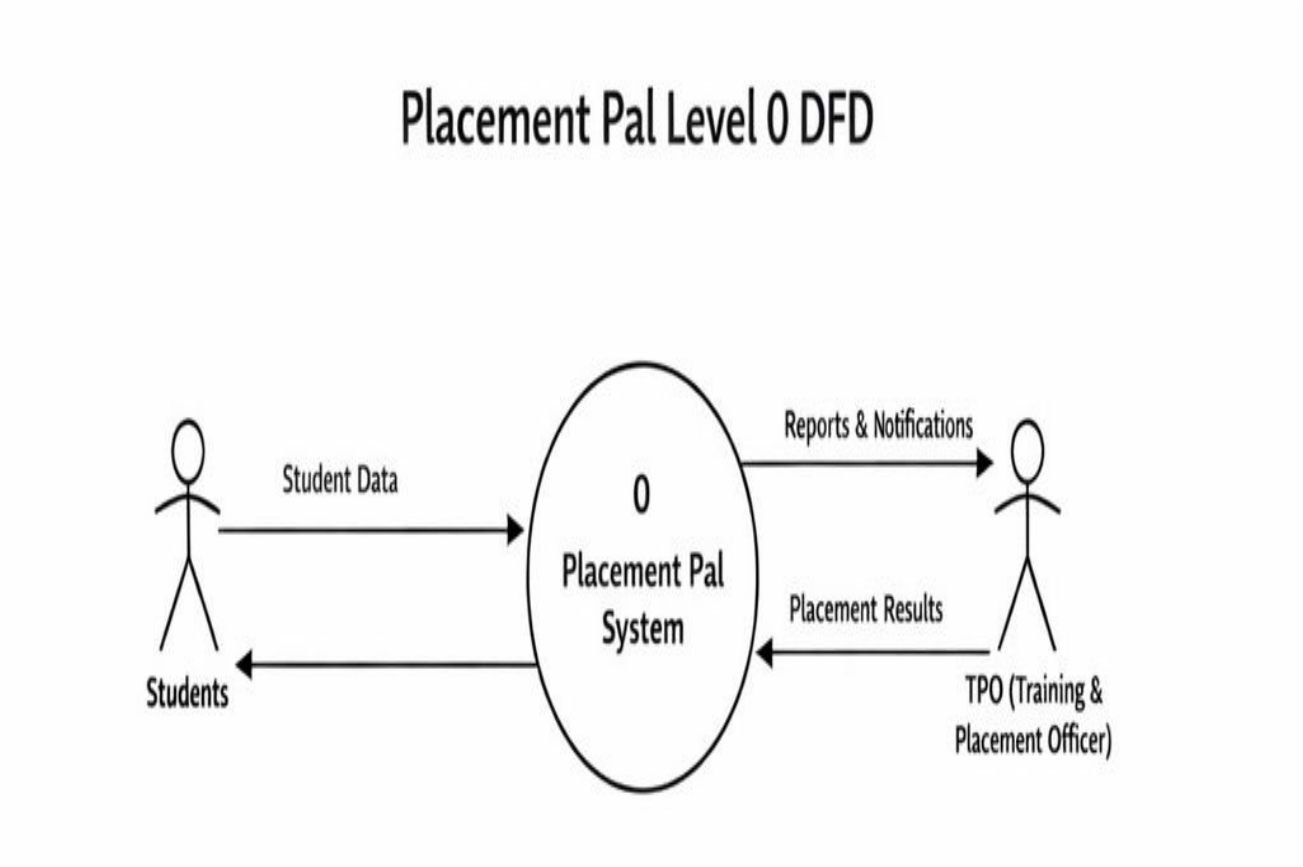

Fig (5.2) Level 0 DFD (Context Diagram)

The Level 0 DFD represents the entire Placement Pal system as a single process.

Students provide input such as student data, assessment responses, and resumes.

The system processes this data and generates outputs like performance reports, learning paths, and notifications.

TPOs interact with the system to receive placement results, reports, and updates.

This diagram provides a high-level overview of how the system interacts with external entities.

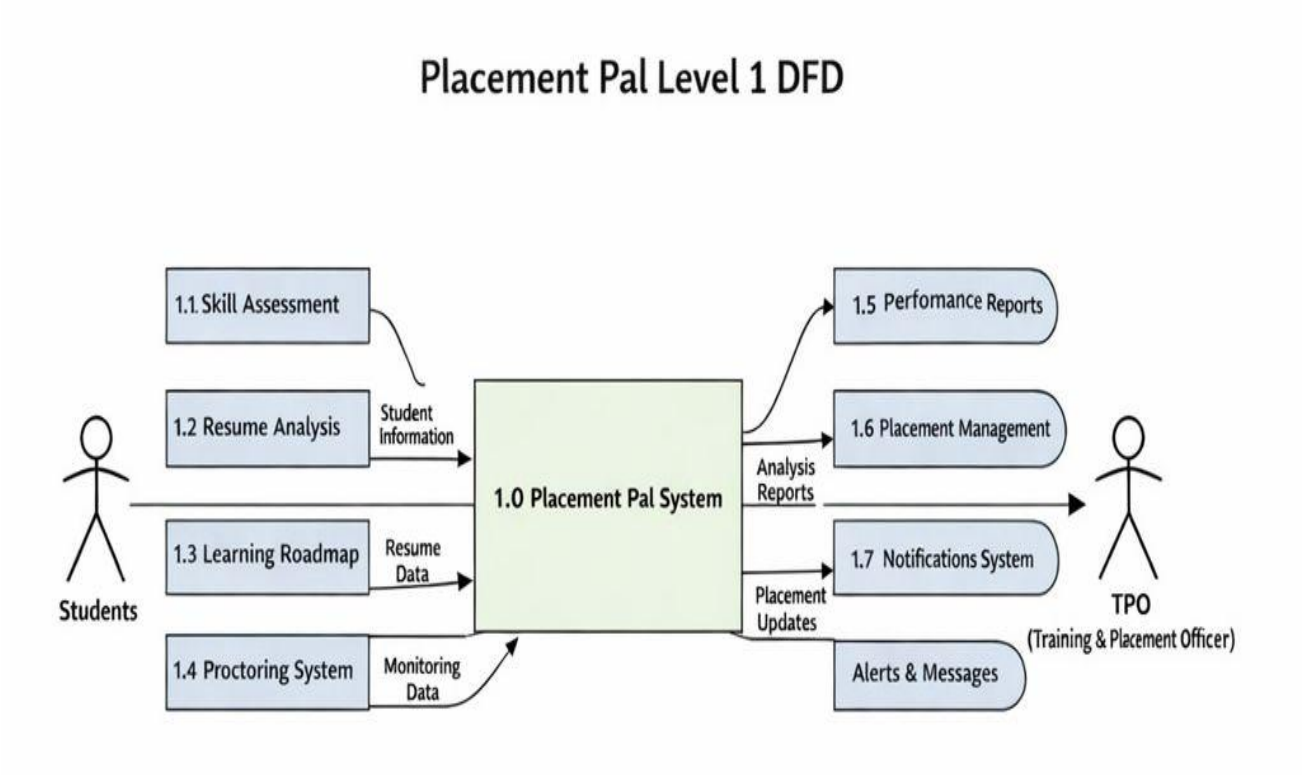

Fig (5.3): Level 1 DFD

The Level 1 DFD expands the system into detailed sub-processes:

Process 1.1 – Skill Assessment:

Students take assessments, and their responses are evaluated to generate performance data.

Process 1.2 – Resume Analysis:

Resumes are analyzed using NLP to extract skills and relevant information.

Process 1.3 – Learning Roadmap:

Based on analysis, the system generates personalized learning paths.

Process 1.4 – Proctoring System:

Monitors student behavior during assessments to ensure fairness.

Process 1.5 – Performance Reports:

Generates detailed reports for both students and TPOs.

Process 1.6 – Placement Management:

Handles placement-related activities such as tracking and updates.

Process 1.7 – Notification System:

Sends alerts and messages regarding assessments and placements.

5.3 System Architecture:

The System Architecture of Placement Pal (Fig 5.4) follows a modular and decoupled design that separates user interaction, application logic, and AI processing into distinct layers. This ensures scalability, flexibility, and efficient system performance.

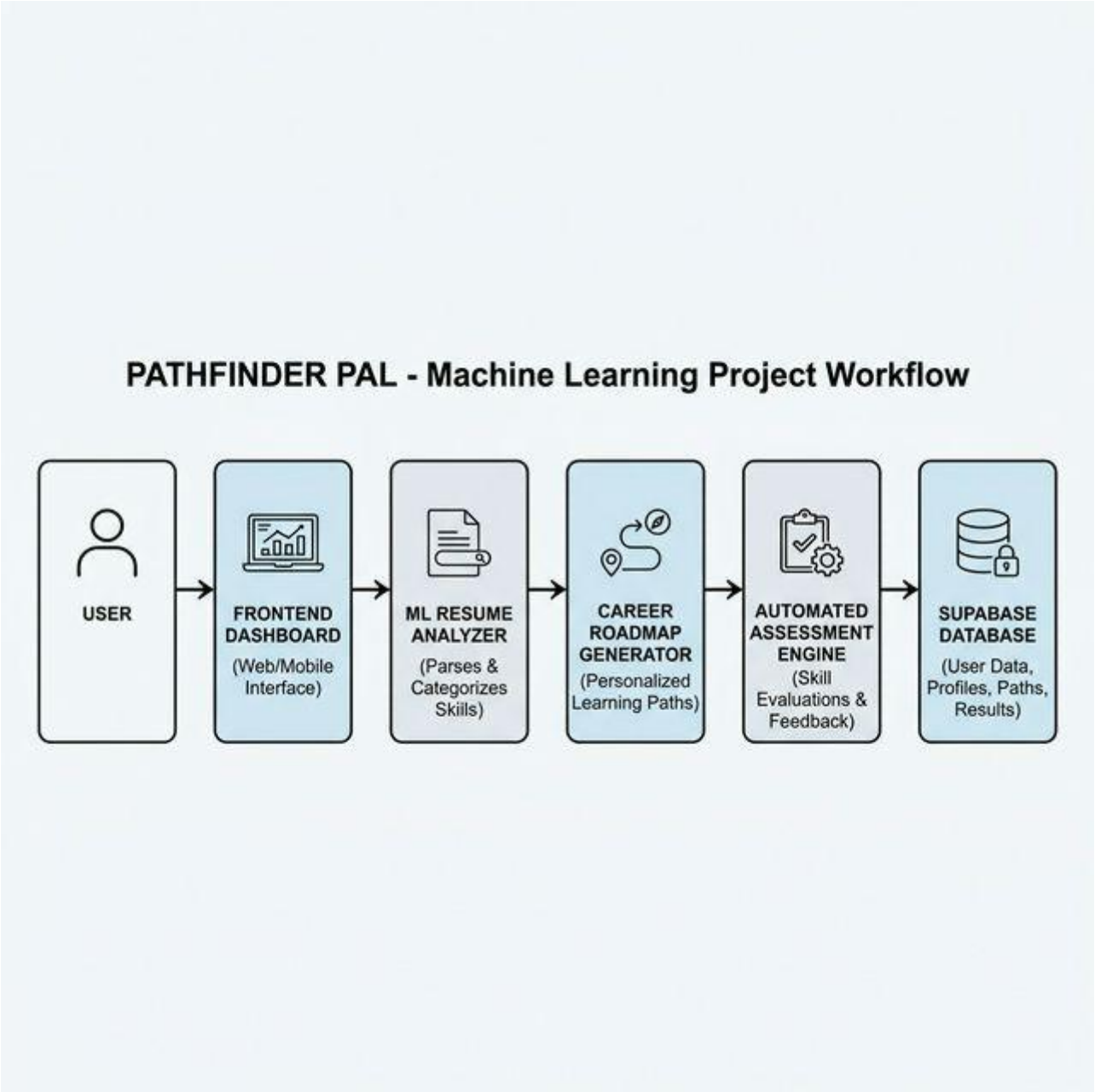

Fig (5.4) Placement Pal System Architecture Diagram

The architecture consists of the following components:

I. User & Frontend Layer (User Interface & Experience):

This layer acts as the interaction point between users and the system, built using React (Vite) and Tailwind CSS.

Provides dashboards, assessments, and learning paths

Displays performance analytics and visualizations

Ensures responsive and user-friendly experience

Communicates with backend via API calls

II. Application & Processing Layer (Business Logic & ML Components):

This layer handles core system functionalities and processing.

ML Resume Analyzer for skill extraction using NLP

Career Roadmap Generator for personalized learning paths

Automated Assessment Engine for evaluation and feedback

Manages performance tracking and readiness prediction

III. Data & Storage Layer (Database & Backend Services):

This layer manages secure data storage and retrieval.

Uses Supabase (PostgreSQL) for storing user data and results

Maintains profiles, learning paths, and performance history

Ensures data consistency, scalability, and real-time access

IV. AI Intelligence Integration:

AI components enhance system capabilities.

Machine Learning for placement readiness prediction

NLP for resume and text analysis

RAG for generating personalized learning content

Supports intelligent assessments and feedback

Conclusion

The architecture ensures efficient processing, scalability, and maintainability by clearly separating system layers. It integrates frontend, backend, and AI components to deliver an intelligent placement preparation platform.

5.4 Implementation:

1) Dashboard (TPO Analytics):

Fig(5.5)

The above figure shows the TPO Analytics Dashboard, which provides batch-wise insights into student performance and skill distribution. It displays key metrics such as total students, domain-wise strengths, and placement readiness. Graphical visualizations like bar charts and pie charts help in understanding skill gaps and trends. This enables TPOs to make data-driven decisions and plan targeted training programs efficiently.

users can choose their preferred language, and a theme toggle option that allows switching between light and dark modes for better user experience.

2) Student Profile Page:

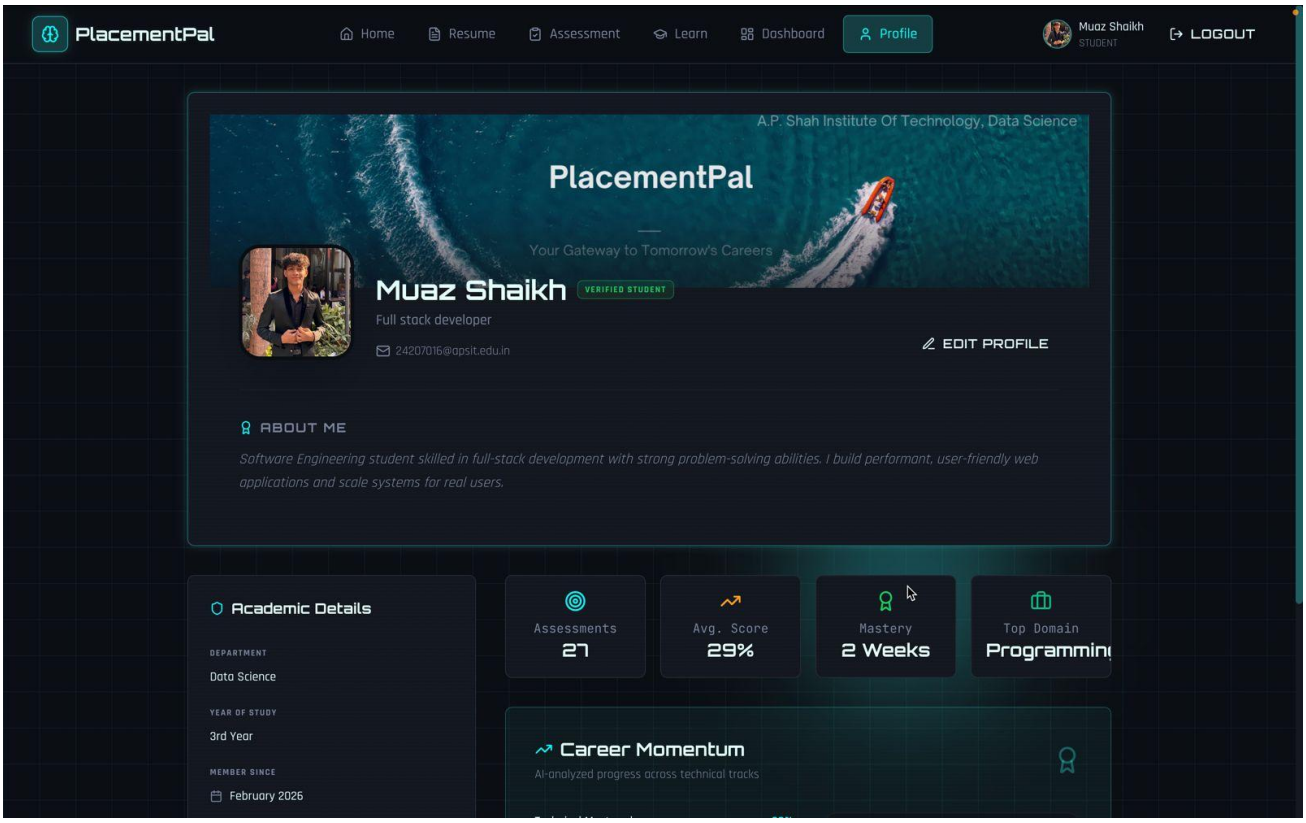

Fig (5.6)

The above figure represents the Student Profile Page, which displays personal details, academic information, and performance metrics. It includes features like assessment count, average score, and mastery level.

The page allows users to update their profile and track their progress. Overall, it provides a comprehensive overview of the student's placement readiness.

3) Student Dashboard:

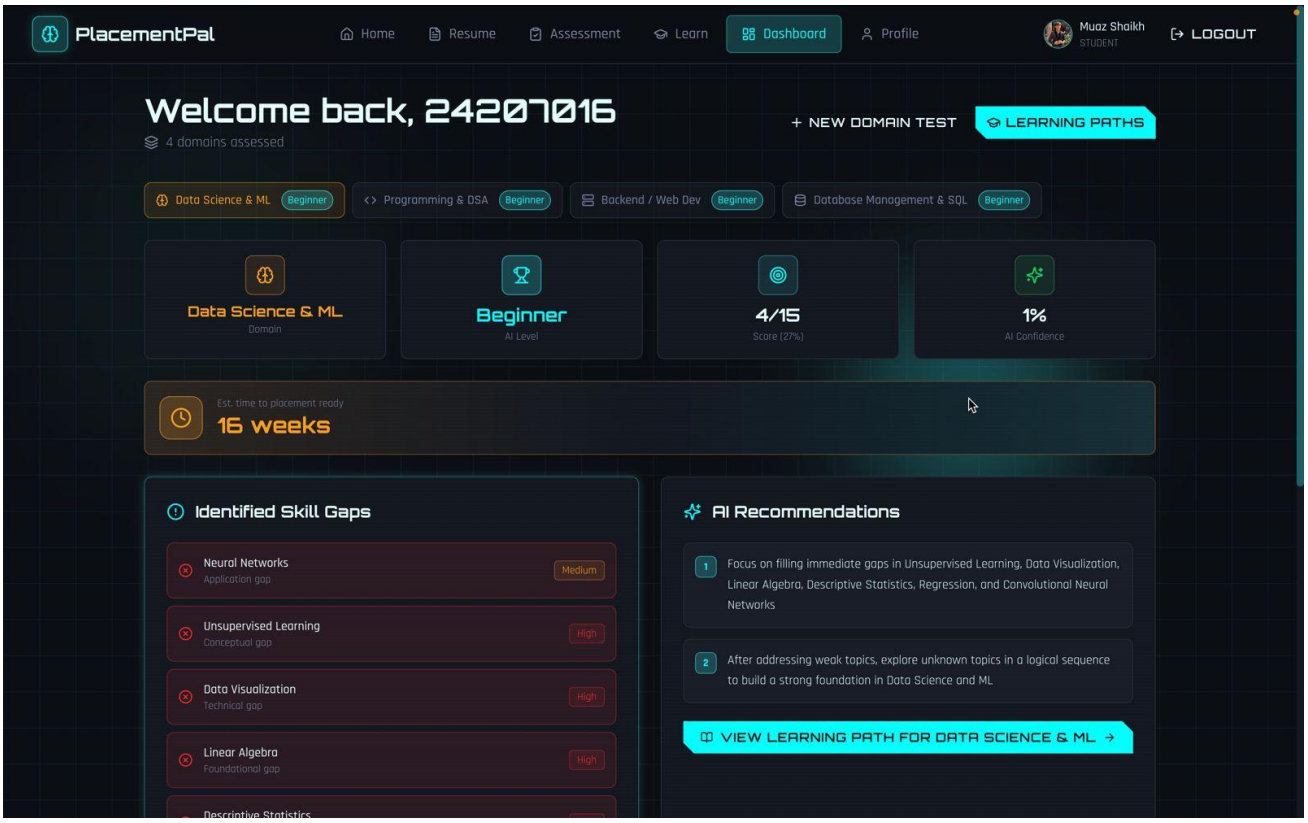

Fig (5.7)

The above figure shows the Student Dashboard, which gives an overview of the student's current progress and performance. It highlights domain levels, scores, and estimated time to become placement-ready.

It also displays identified skill gaps and AI-generated recommendations. This helps students focus on improvement areas effectively.

4) Assessment Module (Track Selection):

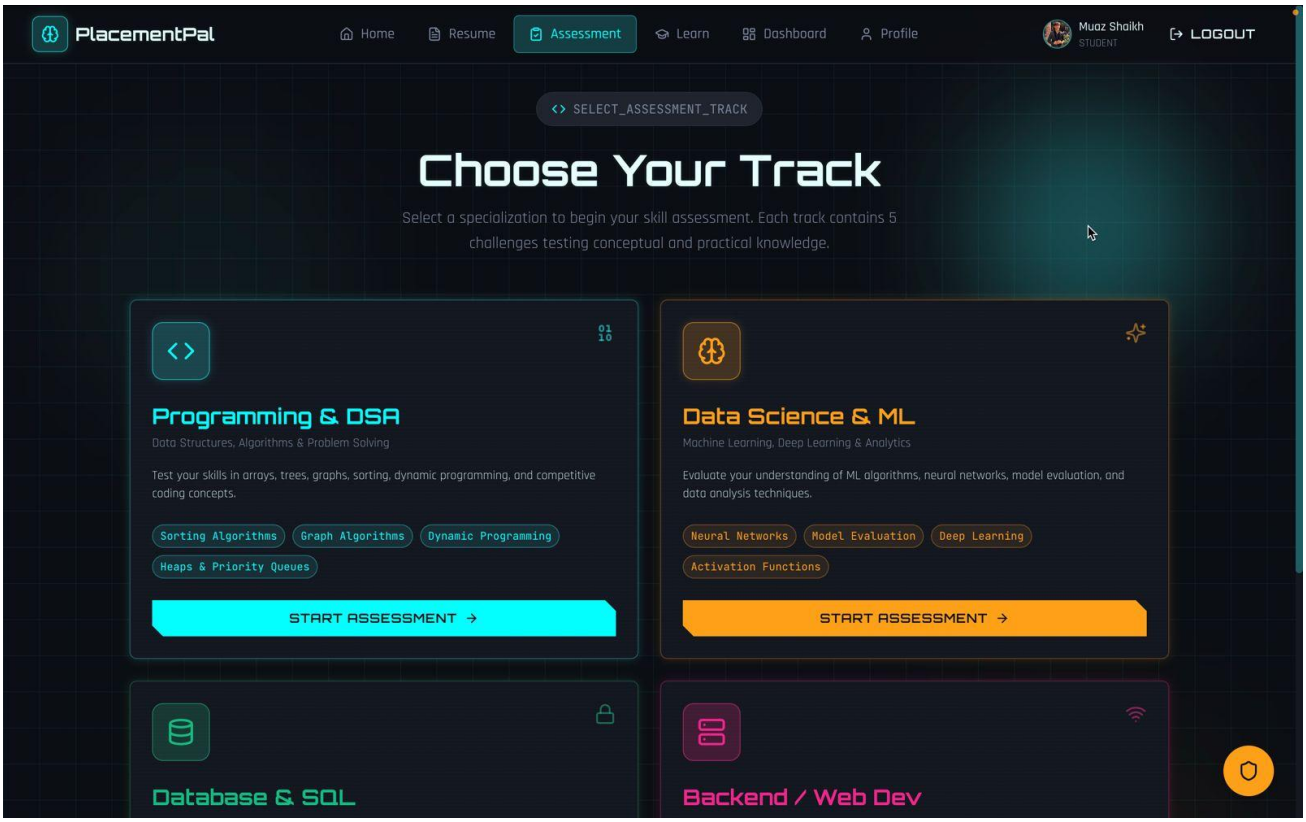

Fig (5.8)

The above figure represents the Assessment Page where students can choose different domains such as Programming, Data Science, Database, and Web Development. Each track includes specific topics and skill areas.

Students can start assessments based on their selected domain to evaluate their knowledge. This ensures domain-specific skill analysis...

5) Learning Path (Personalized Roadmap):

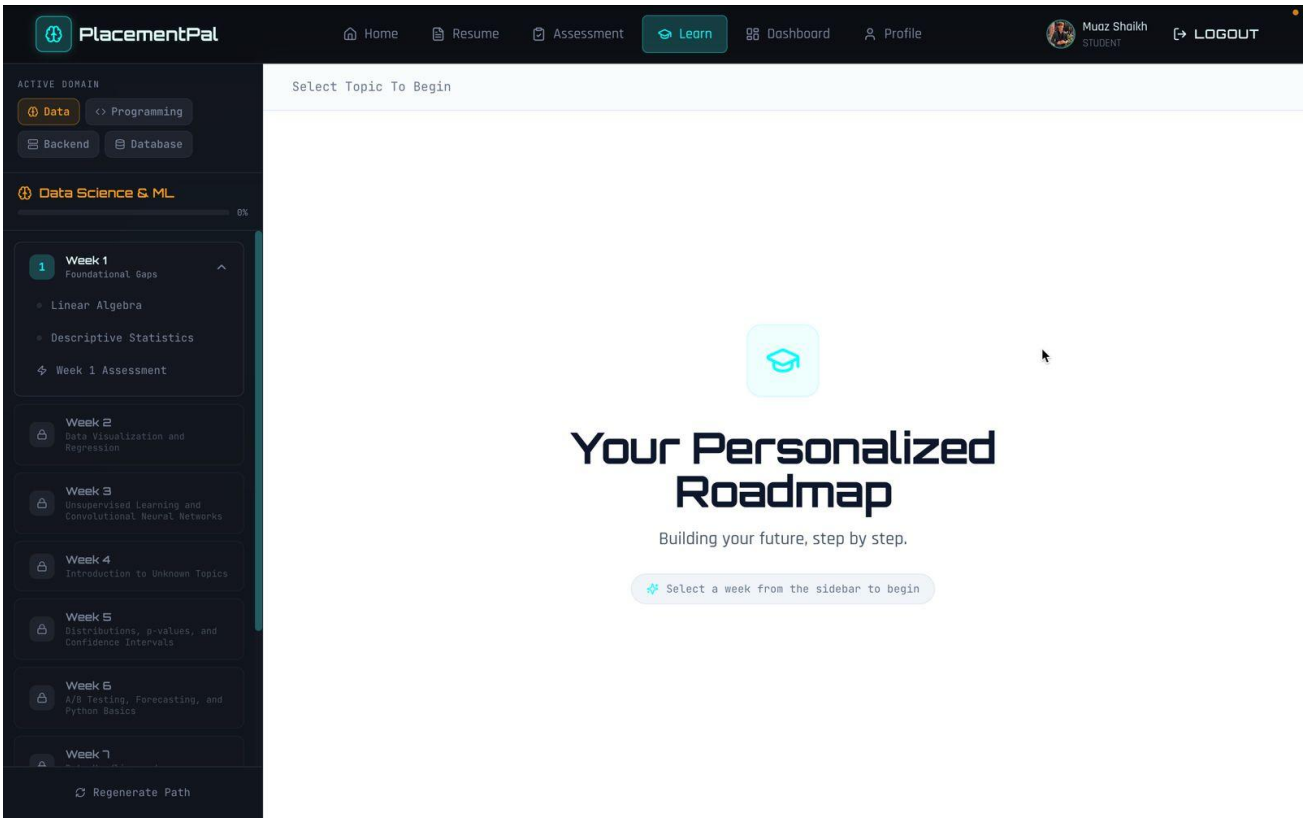

Fig (5.9)

The above figure shows the Personalized Learning Roadmap, which provides a structured weekly plan based on identified skill gaps.

It includes topics, assessments, and progression tracking. The roadmap adapts dynamically using AI to guide students step-by-step. This helps in systematic and efficient preparation.

6) Resume Analysis Module:

Fig (5.10)

The above figure represents the Resume Analysis Page, where students upload resumes and select a target domain. The system analyzes skills using NLP and provides feedback on strengths and improvements.

It helps align the resume with industry requirements. This improves the chances of selection during placements.

7) Home Page:

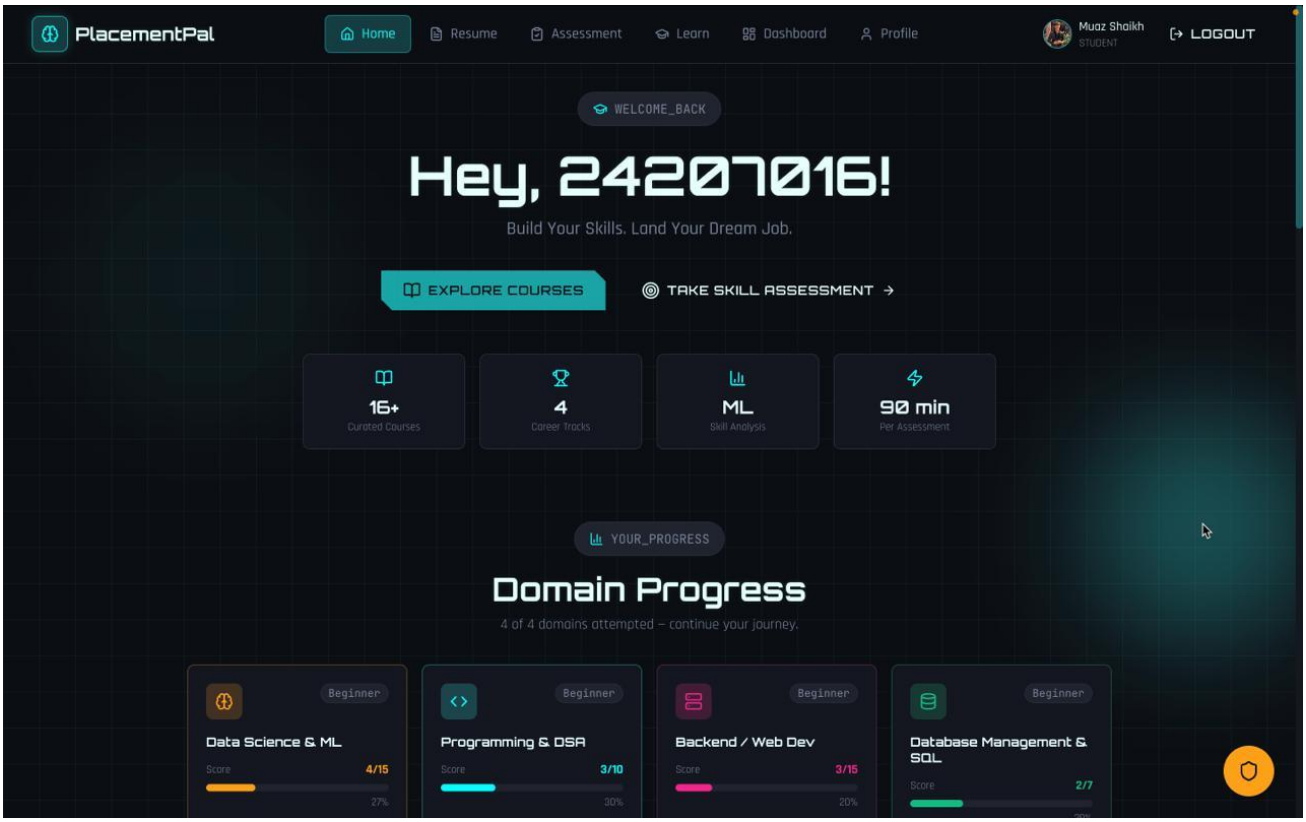

Fig (5.11)

The above figure shows the Home Page of Placement Pal, which acts as the entry point of the platform. It provides quick access to features like courses, assessments, and progress tracking. Key statistics such as total courses and skill analysis are displayed.

The page ensures a smooth and user-friendly navigation experience.

Chapter 6

Technical Specification

Placement Pal uses a combination of frontend, backend, and AI technologies to build a responsive and intelligent system. This stack enables seamless user interaction, data processing, and predictive analysis.

Frontend:

• React (Vite) for building a fast and responsive user interface
• Tailwind CSS for styling and modern UI design
• JavaScript for interactivity and dynamic content handling

Backend:

• Node.js for server-side processing
• Express.js for handling APIs and business logic
• Supabase (PostgreSQL) for database management and authentication

Machine Learning:

• XGBoost / ML models for placement readiness prediction
• Skill gap analysis based on student performance data

NLP Techniques:

• Resume parsing and skill extraction
• Text preprocessing and semantic analysis

AI Technologies:

• Large Language Models (LLMs) for assessment generation and feedback
• Retrieval-Augmented Generation (RAG) for personalized learning roadmaps

Visualization:

• Recharts / Chart libraries for graphs and analytics dashboards

Algorithm Used

Placement Readiness Prediction (Machine Learning Model)

The system uses Machine Learning algorithms (such as XGBoost or classification models) to predict student placement readiness based on performance data, skills, and assessment results.

Chapter 7

Project Scheduling

The project follows a phased Software Development Life Cycle (SDLC) approach, carried out across the academic timeline from January to April 2026. The development process was divided into well-defined stages including topic finalization, requirement analysis, system design, frontend and backend development, AI model integration, and testing, ensuring structured progress at each phase. The schedule highlights key milestones, timelines, and team contributions, reflecting a collaborative and systematic approach to building a scalable and intelligent placement preparation platform.

Sr. No.	Group Members	Duration	Task Performed
1	All Members	2nd Week of January	Group formation, topic finalization, and identification of scope and objectives.
2	All Members	3rd Week of January	Identification of core functionalities and feature requirements.
3	Member 1 & 2	1st-2nd Week of February	Design of user interface (UI) and frontend architecture using React and Tailwind CSS.
4	Member 3 & 4	3rd-4th Week of February	Dataset preparation including student performance data, assessment modules, and skill mapping.
5	Member 2 & 3	1st-2nd Week of March	Backend development using Node.js and Express, including API creation, authentication, and database setup (Supabase/PostgreSQL).
6	Member 1 & 4	3rd Week of March	AI integration including Machine Learning models, NLP-based resume analysis, and RAG-based roadmap generation.
7	All Members	4th Week of March	AI integration including Machine Learning models, NLP-based resume analysis, and RAG-based roadmap generation.

GANTT CHART TEMPLATE

A Gantt chart's visual timeline allows you to
Smartsheet Tip - see details about each task as well as project dependencies.

PROJECT TITLE: Placement Pal

PROJECT GUIDE: Prof. Shubhangi Soni

INSTUTUTE & DEPARTMENT NAI AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

DATE: 4-3-26 10/8/24

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	Sakshi Salve, Mohit Kadam, Dhruvraj Wankhede, Muaz Shaikh	1/8/26	1/16/26	1	100%
1.2	Identifying the functionalities of the Mini Project	Muaz Shaikh, Dhruvraj Wankhede	1/16/26	1/28/26	2	100%
1.3	Discussing the project topic with the help of paper prototype.	Sakshi Salve, Muaz Shaikh	1/28/26	2/5/26	1	100%
1.4	Designing the Graphical User Interface(GUI)	Muaz Shaikh, Dhruvraj Wankhede, Mohit Kadam	2/5/26	2/11/26	1	100%
1.5	Presentation I	Sakshi Salve, Mohit Kadam, Dhruvraj Wankhede, Muaz Shaikh	3/6/26	3/6/26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	Muaz Shaikh, Sakshi Salve	2/17/26	3/3/26	2	100%
2.2	Database Connectivity of all modules	Muaz Shaikh	3/3/26	3/13/26	2	100%
2.3	Integration of all modules and Report Writing	Dhruvraj Wankhede, Mohit Kadam	3/13/26	3/31/26	3	100%
2.4	Presentation II	Dhruvraj Wankhede, Mohit Kadam, Sakshi Salve, Muaz Shaikh	3/31/26	4/1/26	1	100%

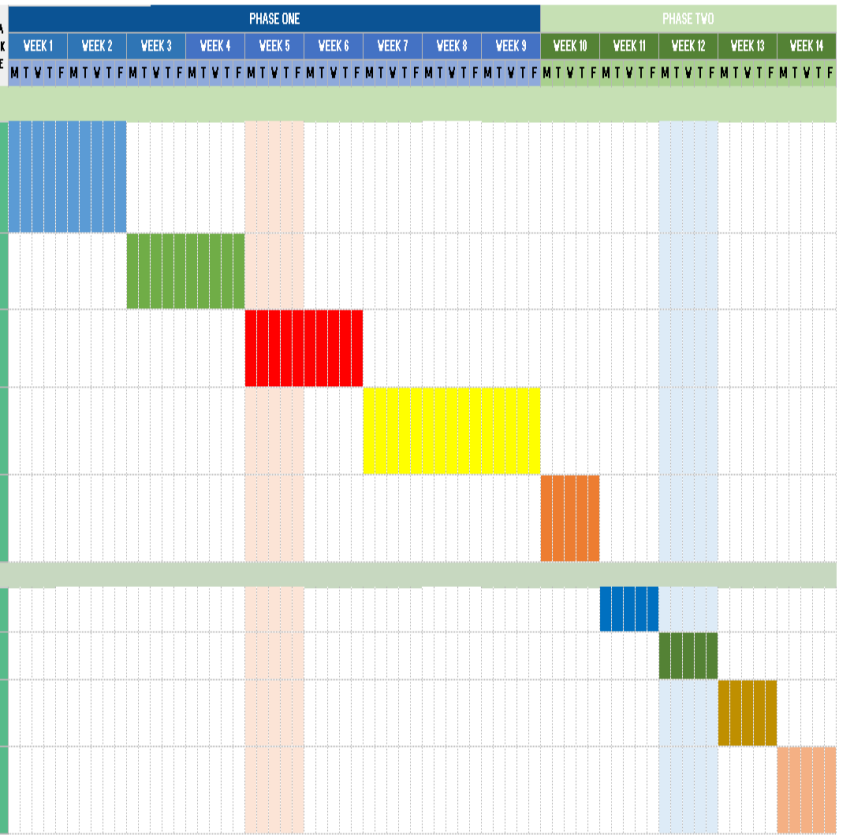

Figure 7.1: Gantt Chart Representing Project Timeline and Task Allocation

Chapter 8

Results

Functional Testing Results:

Extensive functional testing was performed on the Placement Pal system to ensure that all modules operate correctly and meet the specified requirements. The system components including authentication, database operations, AI-based recommendations, and user interface were thoroughly tested.

JWT-based authentication was validated to ensure secure user login and session management. The database system (SQLite/Supabase) was tested for accurate data storage and retrieval. The AI module successfully generated relevant placement recommendations and learning paths based on student data. The dashboards for both students and TPOs were tested for proper data visualization and smooth navigation.

All system modules functioned correctly without major errors, confirming that the system meets its functional requirements.

Feature Accuracy Table

Test Scenario	Expected Result	Observed Result	Status
AI Recommendation System	Relevant placement suggestions	Accurate and relevant recommendations generated	PASS
Authentication (JWT)	Secure login and access control	Unauthorized access successfully blocked	PASS
Password Security (bcrypt)	Secure password storage	All passwords stored in hashed format	PASS
Database Operations	Correct CRUD operations	Data stored and retrieved accurately	PASS
XSS Protection	No script execution	All malicious inputs detected and	PASS

Test Scenario	Expected Result	Observed Result	Status
		blocked	
SQL Injection Prevention	No unauthorized database access	Injection attempts prevented successfully	PASS

Performance Results

Metric	Target Value	Observed Result	Analysis
API Response Time	< 100 ms	42 ms	Fast response due to optimized backend
Database Query Speed	< 80 ms	18 ms	Efficient data retrieval performance
AI Processing Time	< 500 ms	210 ms	Quick ML inference and recommendations
File Upload Processing	< 3 sec	2.8 sec	Smooth handling of resume uploads

The system achieved performance targets with efficient response times and minimal latency, ensuring a smooth user experience

Security Validation

The system was tested against common security threats to ensure reliability and data protection.

•JWT authentication ensured secure session handling and prevented unauthorized access

•Passwords were encrypted using bcrypt, ensuring no plaintext storage

•XSS attacks were prevented through input sanitization

•SQL injection attacks were blocked using parameterized queries

• Rate limiting was implemented to prevent brute-force attacks

All security measures were successfully validated, confirming the system's robustness

Usage Analysis

Usage analysis shows that the platform is widely adopted in urban and semi-urban areas. The majority of users belong to metro (40%) and city (30%) regions, indicating strong adoption in digitally advanced areas.

Town users contribute around 20%, showing good reach in semi-urban regions. However, lower usage in rural (4%) and village (6%) areas highlights the need for improved accessibility and awareness.

Overall, the system performs effectively across different user groups and demonstrates potential for further expansion.

The results confirm that Placement Pal is a reliable, efficient, and secure platform for placement preparation. The system successfully meets functional, performance, and security requirements while providing accurate insights and recommendations.

Chapter 9

Conclusion

Placement Pal demonstrates how Machine Learning, Natural Language Processing (NLP), and Generative AI can be effectively applied to improve placement preparation and student performance analysis. The system successfully automates the process of skill evaluation, readiness prediction, and personalized learning, reducing manual effort and enhancing efficiency.

The project highlights the importance of performance analysis, data-driven insights, and intelligent content generation in achieving accurate and meaningful results. The integration of features such as resume analysis, adaptive assessments, and personalized roadmaps further enhances the overall user experience.

Overall, Placement Pal serves as a practical example of how AI can be used in the education and placement domain to bridge the gap between academic learning and industry requirements. It also provides a strong foundation for future enhancements and advanced applications in intelligent career guidance system

Chapter 10

Future Scope

The Placement Pal system offers significant potential for future enhancements to improve its capabilities, scalability, and real-world impact:

• Integration with Real-Time Job Portals:

The system can be integrated with job and placement APIs (such as LinkedIn, Internshala, or company portals) to provide live updates on job openings, internships, and campus drives. This will enable students to apply directly through the platform and stay updated with relevant opportunities.

• Advanced Deep Learning Models:

Implementation of advanced models such as Neural Networks, Transformers, or fine-tuned LLMs can improve the accuracy of placement readiness prediction and skill analysis. These models can better capture complex relationships between skills, performance, and job requirements.

• Multilingual Support:

Adding support for multiple regional and international languages will make the platform more accessible to a diverse range of users. This can be achieved using multilingual NLP models, enabling students from different backgrounds to interact with the system comfortably.

• Personalized Career Recommendations:

The system can be enhanced to provide detailed career guidance by analyzing user preferences, performance trends, and industry demands. It can recommend specific job roles, certifications, and career paths tailored to individual profiles.

• Mobile Application Development:

Developing a dedicated mobile application (Android/iOS) will improve accessibility and user engagement. Features such as push notifications, offline access, and mobile-friendly assessments will enhance usability.

• AI-Based Interview Simulation:

Integration of mock interview systems using AI can help students practice technical and HR interviews. Features such as real-time feedback, speech analysis, and confidence evaluation can significantly improve interview performance.

• Advanced Analytics & Recruiter Dashboard:

The platform can include enhanced dashboards for recruiters, providing insights into candidate performance, skill distribution, and hiring trends. This will help recruiters make informed hiring decisions efficiently.

• Continuous Learning & Model Improvement:

The system can incorporate feedback loops and real-time data to continuously retrain and improve ML models. This ensures that predictions and recommendations remain accurate and up-to-date with changing industry requirements.

• Integration with Learning Platforms:

Future versions can connect with platforms like Coursera, Udemy, or YouTube to provide direct access to recommended learning resources, making the roadmap more actionable.

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